

Billingsley's Probability Measure Axiom and Collective Exhaustion

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Classical starting point

In Billingsley's formulation, a probability measure is a set function P on a field $\mathcal{F}$ of subsets of Ω satisfying:

- (i) $P(A) \geq 0$ for every $A \in \mathcal{F}$;
- (ii) $P(\Omega) = 1$;
- (iii) if $A_1, A_2, \dots \in \mathcal{F}$ are pairwise disjoint and $\bigcup_n A_n \in \mathcal{F}$, then

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

For a finitely additive probability, this third axiom is equivalent to continuity at the empty set:

$$E_1 \supseteq E_2 \supseteq \dots, \quad \bigcap_n E_n = \emptyset \implies P(E_n) \rightarrow 0.$$

Thus the analytic content of countable additivity is that mass cannot persist along a descending sequence of events that has vanished.

Paper I starting point

Paper I begins one step earlier. Instead of assuming a single countably additive probability measure, it starts with a directed query system: Boolean algebras $\mathcal{E}_i$ connected by refinement maps π_{ij} , together with compatible finitely additive charges $\{\ell_i\}$.

The structural input is a query system consisting of

$$(\mathcal{I}, \leq), \quad (O_i, \mathcal{E}_i)_{i \in \mathcal{I}}, \quad \pi_{ij} : O_j \rightarrow O_i \quad (i \leq j).$$

These data are required to satisfy the standing refinement identities

$$\pi_{ii} = \text{id}, \quad \pi_{ik} = \pi_{ij} \circ \pi_{jk} \quad (i \leq j \leq k),$$

and the relevant directedness hypotheses used by the chosen extension route. No separate underlying sample space is needed at this stage. When a realised space is required, Paper I may take the canonical instantiation

$$\Omega = \varprojlim_{i \in \mathcal{I}} O_i = \left\{ (\omega_i)_i \in \prod_i O_i : \pi_{ij}(\omega_j) = \omega_i \text{ for } i \leq j \right\},$$

with evaluation maps $\text{eval}_i(\omega) = \omega_i$. Thus

$$\text{eval}_i = \pi_{ij} \circ \text{eval}_j \quad (i \leq j)$$

holds by construction. The additional evaluation and separation hypotheses used in the descent to a realised observable measure are:

$$\text{eval}_i : \Omega \rightarrow O_i \text{ is surjective,}$$

and, for the Stone route, discriminability:

$$\omega \neq \omega' \implies \exists i \in \mathcal{I}, \exists A \in \mathcal{E}_i \text{ such that } \text{eval}_i(\omega) \in A, \quad \text{eval}_i(\omega') \notin A.$$

The observable cylinder algebra is

$$\mathcal{C} = \{\text{eval}_i^{-1}(A) : i \in \mathcal{I}, A \in \mathcal{E}_i\}.$$

The observable-data analogue of Billingsley's definition is not yet a probability measure on a σ -algebra. It is an admissible coherent probability datum: a family $\{\ell_i\}$ satisfying the following conditions.

- (i) Local finite probability: each ℓ_i is a normalised finitely additive charge on $\mathcal{E}_i$, i.e.

$$\ell_i(A) \geq 0 \quad (A \in \mathcal{E}_i), \quad \ell_i(O_i) = 1,$$

and for disjoint $A, B \in \mathcal{E}_i$,

$$\ell_i(A \cup B) = \ell_i(A) + \ell_i(B);$$

- (ii) Compatibility under refinement:

$$\ell_j(\pi_{ij}^{-1}A) = \ell_i(A) \quad (i \leq j, A \in \mathcal{E}_i);$$

- (iii) Collective exhaustion (CE): if $i \in \mathcal{I}$ and $E_1, E_2, \dots \in \mathcal{E}_i$ satisfy

$$E_1 \supseteq E_2 \supseteq \dots, \quad \bigcap_{n=1}^{\infty} E_n = \emptyset,$$

then there exists $j \geq i$ such that

$$\ell_j(\pi_{ij}^{-1}(E_n)) \longrightarrow 0.$$

Here local finite probability expands into positivity, normalization, and finite additivity at each level. Compatibility is the new query-system coherence condition. CE replaces direct assumption of countable additivity at the observable-data level: no compatible charge family may assign persistent mass to finite observable conditions that collectively vanish under refinement.

These three conditions are the minimal measure-side hypotheses. Together with the structural query-system hypotheses above, they imply

$$\begin{aligned} \exists! P : \sigma(\mathcal{C}) &\rightarrow [0, 1], \\ P &\text{ is countably additive,} \\ (\text{eval}_i)_\# P &= \ell_i \quad (i \in \mathcal{I}). \end{aligned}$$

Thus Paper I does not alter the target notion of probability measure. The delivered object is an ordinary countably additive probability measure on $(\Omega, \sigma(\mathcal{C}))$. The change is in the input: countable additivity is not assumed of a global P at the outset; it is recovered from local finite probability, refinement compatibility, and CE.